Exp No: 27

BINARY SEMAPHORE IN SYSTEMC

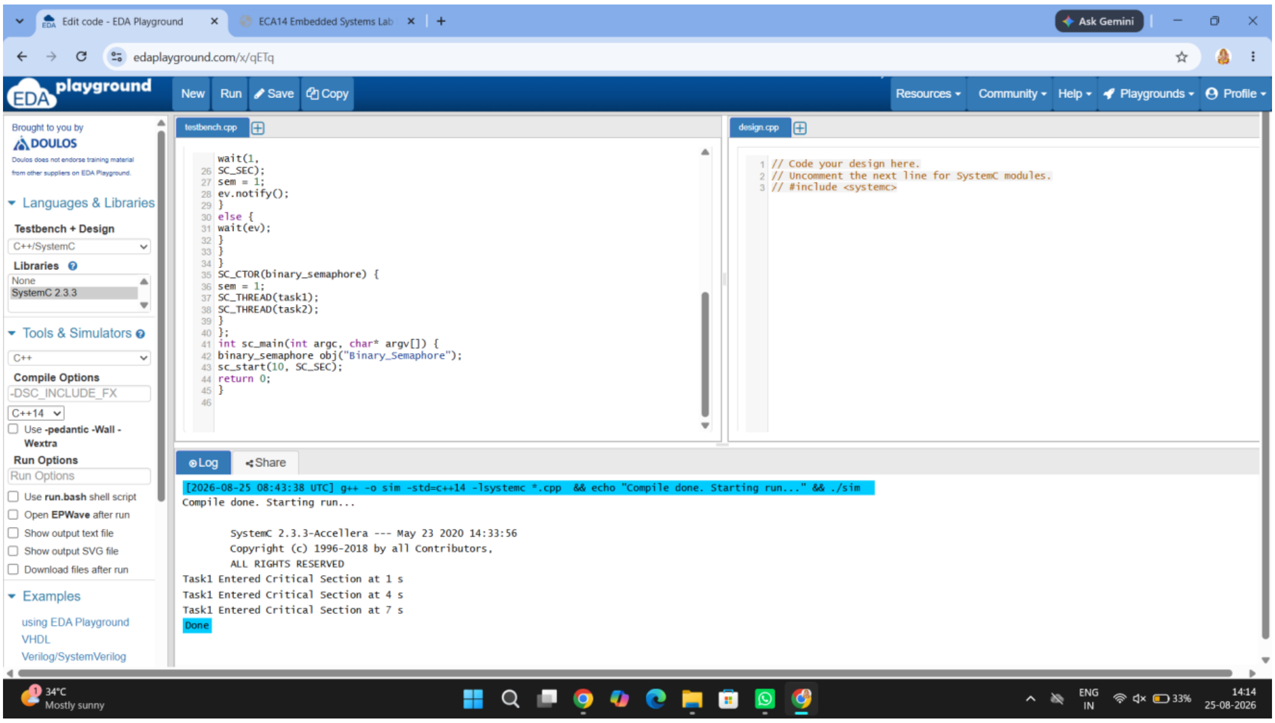
AIM:

To implement and simulate a Binary Semaphore using SystemC for task synchronization.

PROGRAM:

```
#include <systemc.h>

SC_MODULE(binary_semaphore) {

 int sem;
 sc_event ev;

 void task1() {
 while(true) {
 wait(1, SC_SEC);

 if(sem == 1) {
 sem = 0;
 cout << "Task1 Entered Critical Section at " << sc_time_stamp() << endl;  wait(2,
SC_SEC);
 sem = 1;
 ev.notify();
 }
 else {
 wait(ev);
 }
 }
 }
```

```
void task2() {
while(true) {
wait(2, SC_SEC);

if(sem == 1) {
sem = 0;
cout << "Task2 Entered Critical Section at " << sc_time_stamp() << endl;  wait(1,
SC_SEC);
sem = 1;
ev.notify();
}
else {
wait(ev);
}
}
}

SC_CTOR(binary_semaphore) {
sem = 1;
SC_THREAD(task1);
SC_THREAD(task2);
}
};

int sc_main(int argc, char* argv[]) {
binary_semaphore obj("Binary_Semaphore");
sc_start(10, SC_SEC);
return 0;
}
```



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Use -pedantic -Wall -Wextra
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Show output text file
Show output SVG file
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Examples
using EDA Playground
VHDL
Verilog/SystemVerilog
testbench.cpp
wait(1,
SC_SEC);
sem = 1;
ev.notify();
}
else {
wait(ev);
}
}
}
SC_CTOR(binary_semaphore) {
sem = 1;
SC_THREAD(task1);
SC_THREAD(task2);
}
};
int sc_main(int argc, char* argv[]) {
binary_semaphore obj("Binary_Semaphore");
sc_start(10, SC_SEC);
return 0;
}
design.cpp
// Code your design here.
// Uncomment the next line for SystemC modules.
// #include <systemc>
Log
Share
[2026-08-25 08:43:38 UTC] g++ -o sim -std=c++14 -lsystemc *.cpp && echo "Compile done. Starting run..." && ./sim
Compile done. Starting run...
SystemC 2.3.3-Accellera --- May 23 2020 14:33:56
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ALL RIGHTS RESERVED
Task1 Entered Critical Section at 1 s
Task1 Entered Critical Section at 4 s
Task1 Entered Critical Section at 7 s
Done
34°C
Mostly sunny
ENG
IN
33%
14:14
25-08-2026